~Commitment, Consensus, and Admissibility: Reframing the Past, Language, and Scientific Change

The History and Future of the Cathedral We Build

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Introduction

The preceding parts have established a framework in which both mathematics and physics arise within domains defined by admissibility. Mathematical systems emerge through commitment and stabilise through consensus, while physical quantities are admitted through measurement interactions and form relational structures that give rise to what are commonly termed constants.

Now situated, a natural extension follows. If the objects and relations of science are governed by admissibility, then the development of scientific knowledge over time must also reflect transitions between admissibility domains. What is commonly described as error, revision, or progress may instead be understood as a reconfiguration of the conditions under which objects, measurements, and explanations are admitted.

This part examines that extension. It considers how historical scientific frameworks may be reinterpreted through the lens of admissibility, how language participates in stabilising such frameworks, and how the past may be reframed in order to clarify the structure of the present.

1. Historical Stability and the Structure of Apparent Error

Scientific history is often presented as a sequence of corrections, in which earlier theories are replaced by more accurate descriptions of reality. Within this narrative, frameworks such as humoral medicine or phlogiston chemistry are treated as examples of error—necessary steps on a path toward truth. However, this characterisation obscures an important structural feature.

These systems were not arbitrary or incoherent. They exhibited internal consistency, explanatory power within their domains, and long-term stability supported by institutional practice. They provided frameworks within which observations could be interpreted, predictions could be made, and interventions could be justified. From the perspective developed in this work, such systems may be understood as *admissibility domains*.

Their objects—whether humours or phlogiston—were not isolated empirical entities in the modern sense, but stabilised constructs within a network of interpretation and measurement.

They were admissible insofar as they could be consistently invoked within the practices that defined the system.

The transition away from such frameworks did not occur solely because they were logically invalid. Rather, it followed from the introduction of measurement procedures and conceptual structures that rendered their core entities unstable. As measurement resolution increased and alternative relational structures became available, the admissibility of these entities could no longer be maintained.

In this light, historical theories are not best understood as incorrect descriptions of a fixed reality, but as stable configurations within particular admissibility conditions. Their eventual displacement reflects a shift in those conditions, rather than a simple replacement of error by truth.

2. Scientific Change and the Limits of Paradigm Description

The dynamics of scientific change have been examined extensively, most notably in the work of Thomas Kuhn. Kuhn introduced the concept of paradigms to describe the frameworks within which scientific activity is conducted, and he argued that transitions between paradigms involve discontinuities that cannot be fully resolved within a single system of reasoning.

This analysis captures an essential feature of scientific development: that different frameworks may operate with distinct standards of admissibility, such that direct comparison becomes difficult or incomplete. However, the paradigm concept remains primarily descriptive. It identifies the presence of shifts, but does not fully articulate the underlying mechanism by which such shifts occur. Within the present framework, these transitions may be understood as changes in admissibility.

A scientific framework defines not only what problems are addressed and what methods are employed, but what entities are considered to exist, what measurements are meaningful, and what forms of explanation are valid. When these conditions change, the admissibility domain itself is reconfigured.

Paradigm shifts, in this sense, are transitions between admissibility structures. They do not merely replace one set of answers with another; they alter the space of possible questions.

3. Language as a Constraint on Admissibility

Language plays a central role in the formation and stabilisation of admissibility domains. Scientific terms are often treated as labels applied to pre-existing entities. Yet in practice, they function as anchors within a conceptual system. A term does not simply denote; it constrains. It defines what may be referred to, how it may be measured, and how it may be related to other entities.

Within a given framework, language and admissibility are tightly coupled. The introduction of a term such as “phlogiston” or “field” establishes not only a name, but a set of permissible operations and interpretations. These terms become embedded in experimental practice, theoretical development, and educational transmission.

As a result, language contributes to the stability of a framework. It reinforces admissibility by providing a structured means of expression through which concepts are reproduced and extended. When admissibility conditions shift, language must also change. Existing terms may be reinterpreted, restricted, or abandoned, while new terms are introduced to reflect revised structures of measurement and relation. This process is not merely terminological. It reflects a transformation in the underlying constraints of the system. Language, in this sense, may be understood as a surface through which admissibility is both expressed and maintained.

4. Reinterpreting the Past as Admissibility Structure

If scientific frameworks are governed by admissibility, then the interpretation of past theories must be reconsidered. Rather than evaluating earlier systems solely in terms of correctness or error, it becomes possible to analyse them as instances of admissibility under different conditions. Their internal coherence, stability, and eventual displacement provide information about the constraints within which they operated. Reframing the past in this way serves a dual purpose.

First, it clarifies why such systems were effective within their domains. Their success is not anomalous, but a consequence of alignment between their admissible objects and the measurement practices available at the time.

Second, it reveals structural continuities across scientific development. Concepts that appear distinct may share underlying roles within different admissibility frameworks. What changes is not only the entities themselves, but the conditions under which they are admitted.

This perspective shifts the role of history. The past is no longer treated as a catalogue of superseded ideas, but as a record of admissibility structures and their transitions. It becomes a source of insight into the mechanisms by which scientific systems stabilise and evolve.

5. The Present as a Historical Configuration

The analysis developed here applies not only to past frameworks, but to the present. Contemporary scientific systems, including those of modern physics, operate within their own domains of admissibility. They are grounded in established measurement procedures, supported by extensive consensus, and reinforced through institutional and educational structures.

These systems exhibit the same features identified in historical examples: internal coherence, predictive success, and stabilisation over time. Their foundational elements—whether constants, fields, or other constructs—are treated as admissible within the prevailing framework.

The possibility arises that some of these elements may occupy a role analogous to that of earlier constructs such as phlogiston. This does not imply error in the conventional sense, nor does it diminish the effectiveness of current theories. Rather, it suggests that certain entities may be stabilised within present admissibility conditions in ways that could be reinterpreted under future frameworks.

To recognise this possibility is not to undermine contemporary science, but to situate it within the broader structure of scientific development. The present, like the past, may be understood as a configuration shaped by admissibility, subject to transformation as those conditions evolve.

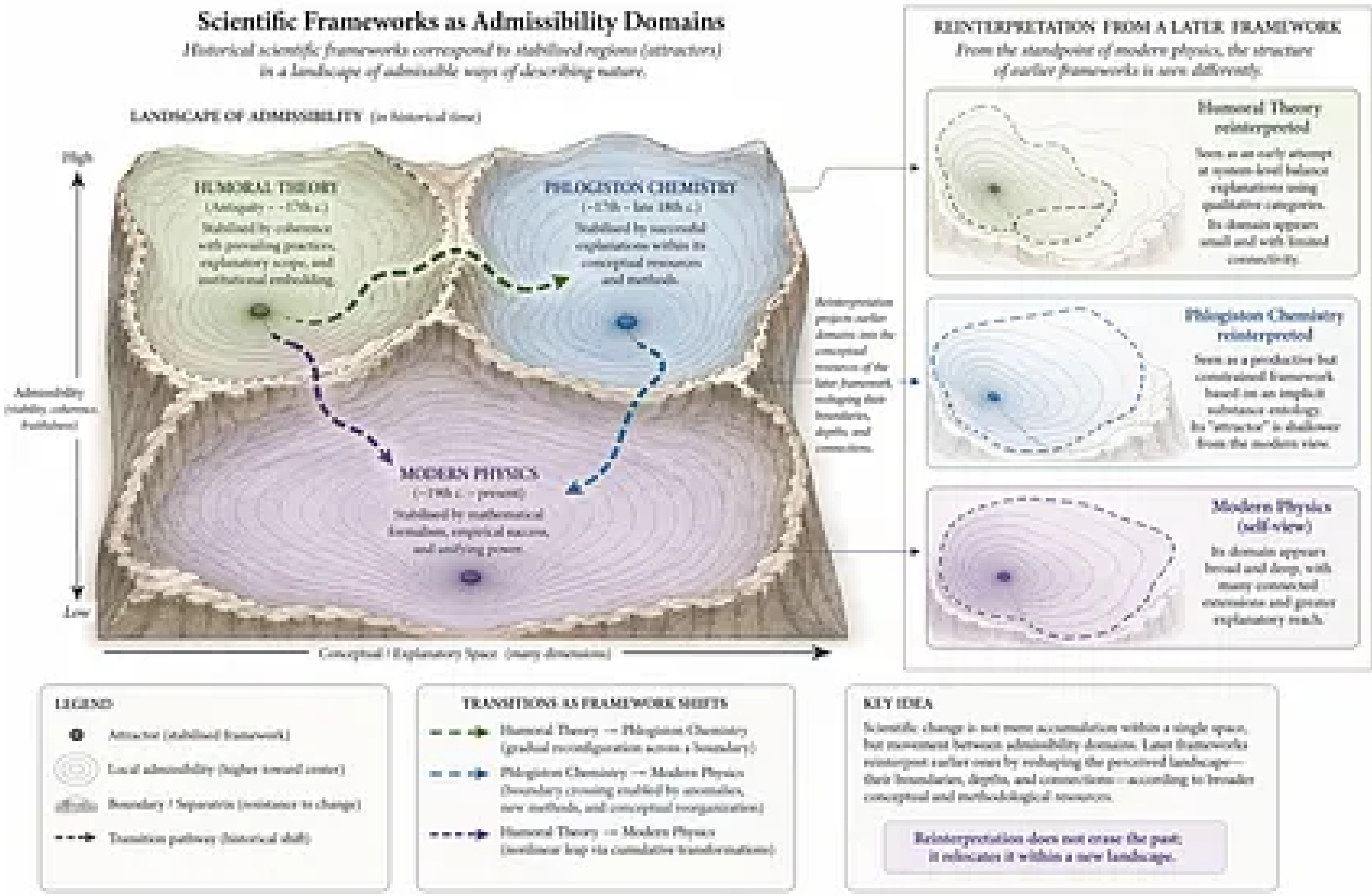
6. Implications for Mathematical and Physical Practice

The perspective developed across these parts has implications for both mathematics and physics. If admissibility is prior to logic and measurement, then the development of new frameworks requires attention to the conditions under which objects and relations are admitted. Innovation does not arise solely from new results within existing systems, but from shifts in the admissibility criteria that define those systems.

Approaches that ground mathematical and physical reasoning in finite, measurable interactions—treating uncertainty as intrinsic and relational structure as primary—may be understood as attempts to establish alternative admissibility domains. Such approaches do not negate existing frameworks, but re-contextualise them within a broader landscape.

The coexistence of multiple admissibility domains becomes possible. Each domain may be internally coherent and effective within its scope, while differing in its foundational commitments and constraints.

The task is therefore not to replace one system with another, but to clarify the boundaries and relations between them. By doing so, it becomes possible to navigate between frameworks, identify their domains of applicability, and explore the conditions under which new structures may emerge.



A conceptual diagram illustrating the reinterpretation of historical scientific frameworks as transitions between admissibility domains. The figure depicts multiple stabilised regions (representing distinct frameworks such as humoral theory, phlogiston chemistry, and modern physics), separated by boundaries or separatrices. Arrows and pathways indicate transitions between these domains. The figure shows how reinterpretation from a later framework reshapes the perceived structure of earlier ones.

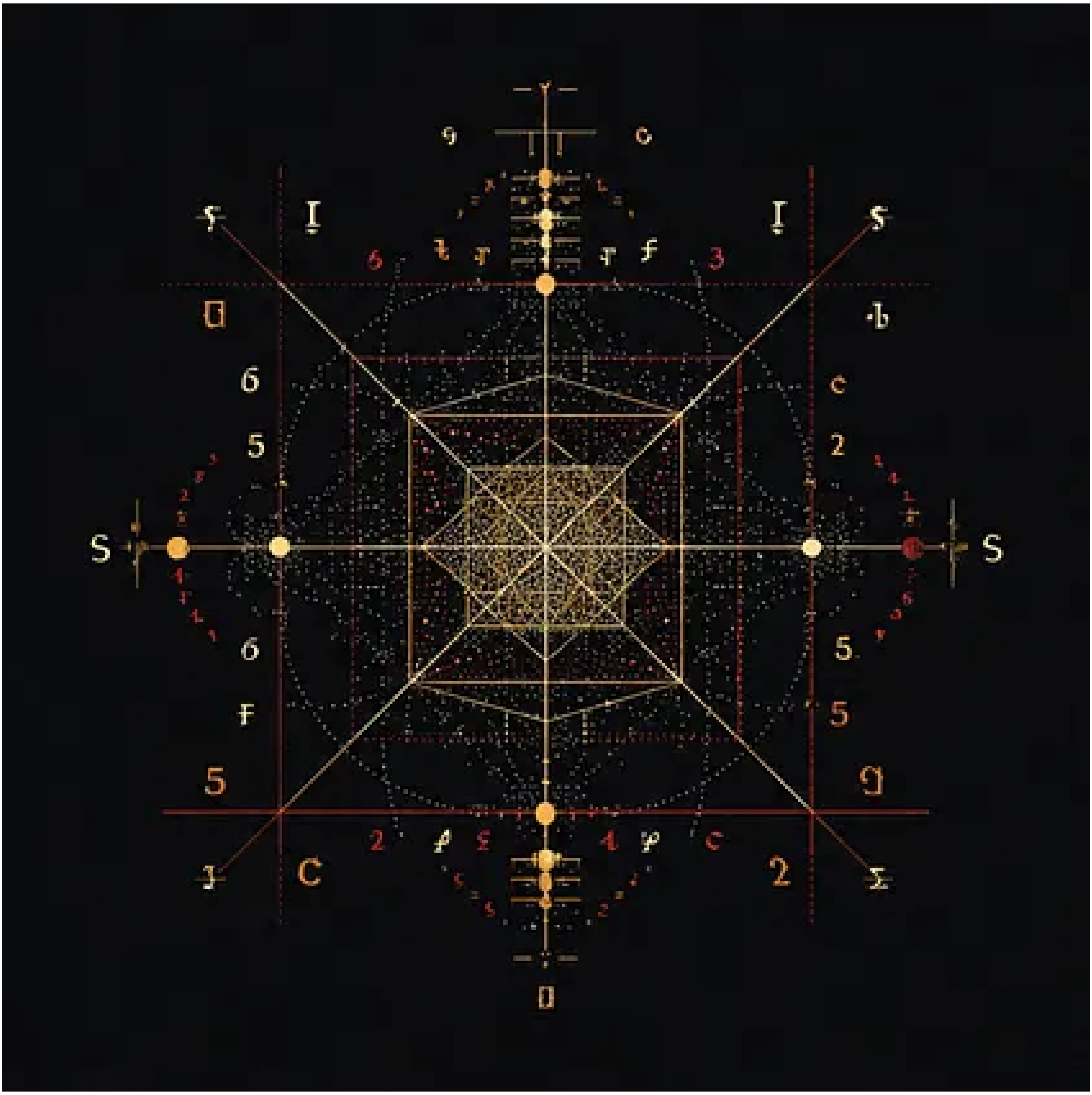
Conclusion: On Seeing the Present Through the Past

The examination of scientific development through admissibility reveals a consistent structure across mathematics, physics, and their histories. Theories arise within domains defined by what may be admitted, stabilise through consensus and practice, and persist until changes in measurement and conceptual structure alter the conditions under which their core entities remain viable.

To understand this process is to recognise that the past is not simply superseded. It is reframed. By interpreting earlier frameworks as admissibility structures rather than as errors, it becomes possible to identify the constraints that shaped them and to observe how similar constraints operate in the present. This reframing does not diminish scientific achievement; it situates it within an evolving structure of understanding.

Clarity, in this context, does not arise from asserting finality, but from recognising the conditions under which knowledge is formed. The present becomes visible not as an endpoint, but as a position within a continuing sequence of admissibility transitions.

In this light, the task of inquiry is not only to extend existing systems, but to examine the boundaries that define them, and to consider what new forms of admissibility may yet emerge.



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